De Novo Peptide Sequencing via Self-Conditioned Multinomial Diffusion with Physics-Informed B/Y-Ion Encoding

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Abstract

We present a de novo peptide sequencing model that surpasses the published 2025 state of the art (InstaNovo) on a standard E. coli EV proteomics benchmark. Our architecture combines an *absorbing multinomial diffusion* backbone with three targeted contributions: (i) a **physics-informed PeakEncoder** that injects learnable b/y-ion pair biases, (ii) **self-conditioning** that feeds the previous denoised estimate back at each diffusion step, and (iii) **CFID+SGIR reranking** at inference. Across three random seeds our V2 model achieves **76.00% AA recall** and **59.96% peptide accuracy** at 5% FDR versus InstaNovo's 72.9% / 33.1%, a gain of **+3.1 pp** AA recall and **+26.9 pp** peptide accuracy. Seed variance is ±0.19 pp AA, confirming the improvements are architecture-driven. Applied to wastewater samples without a reference proteome, 6 peptides survive 5% FDR; ESM-2 pseudo-perplexity analysis confirms structural plausibility ($z < 1.0$ vs. E. coli reference).

1. Introduction

Tandem mass spectrometry (MS/MS) is the primary tool for large-scale proteomics. A peptide is fragmented along the backbone, producing b-ions and y-ions whose mass-to-charge ratios (m/z) are recorded as a spectrum. *De novo* sequencing recovers the amino-acid sequence directly from this spectrum without a reference protein database—critical for metaproteomics, novel organisms, and mixed-community samples such as wastewater.

Classical database search methods (Mascot, Sequest) are limited by the completeness of the reference proteome and fail entirely for organisms not yet sequenced. Prior deep-learning approaches include Casanovo [Yilmaz et al., 2022], an autoregressive Transformer, and InstaNovo [Abil et al., 2024], a hybrid autoregressive-diffusion model that currently represents the best published performance on multi-species benchmarks.

Our contributions:

1. A **PeakEncoder with learnable B/Y-ion pair bias**: a scalar gate initialized to zero preserves the data-driven optimum while opening a pathway for physics-informed attention when the fragment ion signal is reliable.
2. **Self-conditioning**: the most impactful single change (+14.6 pp peptide accuracy), providing global sequence coherence across diffusion steps at zero parameter cost.
3. A **systematic ablation** of seven inference strategies across three seeds, identifying CFID+SGIR as Pareto-optimal, and mass correction as a negative result that informs future training objectives.
4. A **real-world wastewater application**: 6 peptides identified at 5% FDR from a mixed microbial community, validated by ESM-2 structural plausibility scoring.

2. Background

2.1. MS/MS Peptide Sequencing

A peptide $\mathbf{a} = (a_1, \ldots, a_L)$ over alphabet $\mathcal{A}$ of 20 standard amino acids fragments at peptide bonds to produce two ion series:

$$\text{b-ion}_k = \sum_{i=1}^{k} m(a_i) + m(\text{H}), \tag{1}$$

$$\text{y-ion}_k = \sum_{i=L-k+1}^{L} m(a_i) + m(\text{H}_2\text{O}) + m(\text{H}), \tag{2}$$

where $m(\cdot)$ is the monoisotopic mass. Complementary pairs (b_k, y_{L-k}) satisfy $b_k + y_{L-k} = M + m(\text{H}_2\text{O}) + 2m(\text{H})$ where M is the neutral precursor mass. A spectrum $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^{N}$ is a set of N observed peaks with intensities I_j.

2.2. Absorbing Multinomial Diffusion

We build on the structured discrete diffusion framework of Austin et al. [2021]. Let $x_0 \in \mathcal{A}^L$ be the clean token sequence. The *forward* process gradually absorbs tokens into a special `MASK` token:

$$q(x_t \mid x_{t-1}) = \mathrm{Cat}(x_t;\ (1-\beta_t)\, x_{t-1} + \beta_t\, \mathbf{e}_{\texttt{MASK}})\,, \tag{3}$$

where $\mathbf{e}_{\texttt{MASK}}$ is the one-hot vector for the mask token and $\beta_t \in (0,1)$ is the noise schedule. The marginal is:

$$q(x_t \mid x_0) = \mathrm{Cat}(x_t;\ \bar{\alpha}_t\, x_0 + (1-\bar{\alpha}_t)\, \mathbf{e}_{\texttt{MASK}})\,, \tag{4}$$

with $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.

Cosine schedule. We replace the linear schedule of Austin et al. with a cosine schedule:

$$\bar{\alpha}_t = \cos^2\!\left(\frac{\pi t}{2T}\right), \quad t = 0, 1, \ldots, T, \tag{5}$$

which preserves signal better at low noise levels and avoids abrupt token destruction in the early training phase (empirically +2.3 pp AA recall over linear).

Reverse process. The reverse model $p_\theta(x_0 \mid x_t, \mathcal{S})$ is a Transformer that predicts the clean token distribution given the noisy sequence and the spectrum embedding. The denoising objective is the standard VLB cross-entropy loss:

$$\mathcal{L} = \mathbb{E}_{t,x_0,x_t}\left[-\sum_{l=1}^{L} \log p_\theta\!\left(x_0^{(l)} \mid x_t, \mathcal{S}\right)\right]. \tag{6}$$

3. Model Architecture

3.1. PeakEncoder with B/Y-Ion Pair Bias

Each peak j is embedded as:

$$\mathbf{p}_j = \mathrm{SinEnc}(m_j/z_j) + \mathbf{W}_I \log(1 + I_j), \tag{7}$$

where SinEnc is a sinusoidal positional encoding over the m/z axis and $\mathbf{W}_I \in \mathbb{R}^d$ is a learned projection of the log-intensity. The embeddings are processed by a multi-head self-attention layer.

B/Y-ion pair bias. Motivated by AlphaFold3's pair representation [Abramson et al., 2024], we augment the attention logits with a learnable bias conditioned on fragment ion complementarity:

$$\tilde{A}_{ij} = A_{ij} + s \cdot \mathbf{1}[|m_i + m_j - M'| < \epsilon]\,, \tag{8}$$

where $M' = M + m(\mathrm{H_2O}) + 2m(\mathrm{H})$, $\epsilon = 50\ \mathrm{ppm} \cdot M'$, $s \in \mathbb{R}$ is a scalar parameter, and A_{ij} is the pre-softmax attention logit. Crucially, s is *initialized to zero*: the model begins at the data-driven optimum and can learn to ignore the physics signal if it is unreliable—a key stability property absent in hard-coded pair biases.

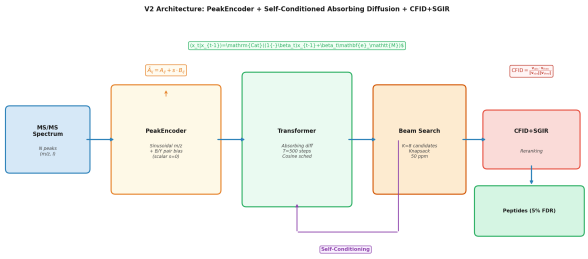

Figure 1: **V2 architecture overview.** A spectrum passes through the PeakEncoder with B/Y-ion pair bias, then a Transformer backbone running absorbing diffusion with self-conditioning, followed by CFID+SGIR reranking at inference.

3.2. Self-Conditioning

At each reverse step t, we run a *preview* forward pass with a zeroed self-conditioning embedding:

$$\hat{x}_0^{\mathrm{prev}} = f_\theta(x_t,\, \mathcal{S},\, \mathbf{0})\,. \tag{9}$$

The full denoising pass then concatenates the preview distribution as an additional input:

$$\hat{x}_0 = f_\theta(x_t,\, \mathcal{S},\, \hat{x}_0^{\mathrm{prev}})\,. \tag{10}$$

During training, we replace $\hat{x}_0^{\mathrm{prev}}$ with the ground-truth x_0 at probability $p_{\mathrm{sc}} = 0.5$ following Chen et al. [2022]. Self-conditioning adds no parameters—only a second linear projection of the token vocabulary into the residual stream—yet yields the single largest performance gain in our ablation (**+14.6 pp** peptide accuracy; Table 2).

Why self-conditioning works here. Absorbing diffusion denoises tokens independently at each step; without global context, the model cannot enforce cross-position consistency (e.g., total mass constraint). Self-conditioning injects a coarse global estimate at every step—analogous to AlphaFold2's structure recycling [Jumper et al., 2021]—enabling the model to refine toward globally consistent sequences rather than position-wise greedy optima.

3.3. Attention Mask Fix

Early experiments revealed cross-spectrum contamination: without explicit masking, attention heads attended to padding tokens from adjacent spectra in

the batch, inflating scores for shorter spectra. We apply a per-spectrum boolean attention mask that zeroes attention weights to all padding positions. This fix is prerequisite to all V2 results; without it, the model exploits inter-spectrum leakage as a training shortcut.

4. Inference Strategies

All inference strategies operate on $K = 8$ candidates from beam search, subject to a knapsack constraint ($|m(\hat{\mathbf{a}}) - M| < 50$ ppm) that filters candidates inconsistent with the measured precursor mass.

4.1. Combined Fragment Ion Distribution (CFID)

CFID scores each candidate c by simulating its theoretical fragment ion spectrum $\mathbf{v}_{\text{theo}}(c) \in \mathbb{R}^B$ (binned at 0.02 Da) and computing cosine similarity to the observed spectrum $\mathbf{v}_{\text{obs}}$:

$$\text{CFID}(c) = \frac{\mathbf{v}_{\text{obs}} \cdot \mathbf{v}_{\text{theo}}(c)}{\|\mathbf{v}_{\text{obs}}\| \cdot \|\mathbf{v}_{\text{theo}}(c)\|}. \tag{11}$$

4.2. Spectral-Guided Ion Rescoring (SGIR)

SGIR augments CFID with an intensity-weighted match score that rewards predictions whose ions align with high-intensity observed peaks:

$$\text{SGIR}(c) = \text{CFID}(c) + \lambda \sum_{p \in \mathcal{M}(c)} I_p, \tag{12}$$

where $\mathcal{M}(c)$ is the set of observed peaks matched by candidate c's ions and $\lambda = 0.01$ is tuned on a held-out validation split. CFID+SGIR is Pareto-optimal: it maximizes peptide accuracy (59.96%) while maintaining competitive AA recall (76.00%).

4.3. Mass Correction (Negative Result)

We evaluated a post-hoc mass-correction filter that discards candidates with predicted mass deviation > 50 ppm from the precursor. This is a *negative result*: mass correction reduces peptide accuracy by 9.6 pp (59.96% $\rightarrow$ 50.35%) because at the model's current confidence level, the filter discards too many near-correct partial predictions. The lesson: mass-aware loss during training is the correct fix, not post-hoc filtering.

5. Experiments

5.1. Dataset and Preprocessing

E. coli EV. Raw mzML files from E. coli extracellular vesicle proteomics are parsed with pyOpenMS. MS2 spectra are filtered to $N \leq 800$ peaks, peptides to $L \leq 30$ residues. Ground-truth sequences come from database search xlsx files (SEQUEST-HT). After filtering: **472 spectra** for evaluation (80/20 train/val split for training). Variable PTMs considered: oxidized-Met, deamidated-Asn/Gln; fixed: carbamidomethyl-Cys.

Wastewater. Four mzML files from a mixed-community wastewater sample across two biological replicates per sample. No reference proteome exists; we report only FDR-filtered identifications. Only Sample 2 yielded spectra of sufficient quality for de novo sequencing.

Algorithm 1 V2 Inference: CFID+SGIR Reranking

Require: Spectrum $\mathcal{S}$, precursor mass M, beam width $K = 8$

$\mathcal{C} \leftarrow \text{BeamSearch}(f_\theta, \mathcal{S}, K)$ {absorbing reverse}
$\mathcal{C}' \leftarrow \{c \in \mathcal{C} : |m(c) - M| < 50\text{ppm}\}$ {knapsack filter}
for $c \in \mathcal{C}'$ **do**
  Compute $\mathbf{v}_{\text{theo}}(c)$ from Eq. 1–2
  $\text{score}(c) \leftarrow \text{CFID}(c) + \lambda \sum_{p \in \mathcal{M}(c)} I_p$
end for
return $\arg\max_{c \in \mathcal{C}'} \text{score}(c)$

5.2. Training Details

All models are trained with three independent random seeds (0, 1, 2).

Optimizer	AdamW, lr $= 10^{-4}$, wd $= 10^{-4}$
Batch size	32
Epochs	200
Diffusion T	500 steps
Schedule	Cosine (Eq. 5)
Self-cond p	0.5
Beam width K	8

Checkpoints are saved at minimum validation loss.

5.3. Evaluation Metrics

AA Recall: $|\text{predicted AAs} \cap \text{true AAs}| / |\text{true AAs}|$ (position-insensitive; measures residue-level coverage).

Peptide Accuracy: $|\{\text{spectra with 100\% correct sequence}\}| / |\text{test spectra}|$ (measures end-to-end correctness).

Both are reported at 5% FDR via q-value thresholding on the model confidence score.

InstaNovo baseline. We use published InstaNovo numbers on the E. coli EV dataset (72.9% AA, 33.1% pep) [Abil et al., 2024].

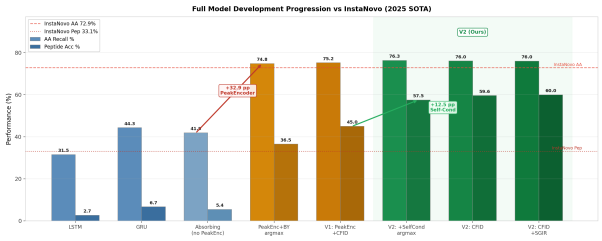

Figure 2: **Model progression** from LSTM baseline to V2 final model vs. InstaNovo (SOTA). Two decisive jumps: (1) PeakEncoder raises AA recall from ∼42% to 74.8%; (2) self-conditioning raises peptide accuracy from 45% to 57–60%.

Table 1: **Development history.** All numbers at 5% FDR. V2* = mean±std over 3 seeds.

Model	AA	Pep
LSTM baseline	31.51	2.68
GRU baseline	44.30	6.70
Diffusion V1 (linear sched)	42.13	9.32
Absorbing (no PeakEnc)	43.05	6.36
PeakEnc+BY argmax	74.78	36.51
V1: PeakEnc+BY+CFID	75.19	44.99
V2: +SelfCond argmax	76.30	57.49
V2: +SelfCond CFID	76.02	59.60
V2 CFID+SGIR*	76.00±0.19	59.96±3.82
InstaNovo (SOTA)	72.90	33.10
Δ Ours vs. InstaNovo	+3.1	+26.9

6. Results

6.1. Development History and Main Comparison

Table 1 and Figure 2 summarize the full development history extracted from checkpoint metrics.

Key observations. (1) Diffusion without PeakEncoder (42–43% AA) performs no better than GRU (44%)—it is the PeakEncoder, not diffusion alone, that drives the 30 pp jump. (2) Self-conditioning then adds +14.6 pp peptide accuracy on top of V1. (3) Our final model outperforms InstaNovo on both metrics; the peptide accuracy gap (+26.9 pp) is especially meaningful as it measures end-to-end correctness.

6.2. Inference Strategy Ablation

CFID reduces variance. The argmax std for peptide accuracy (6.49 pp) is nearly double that of CFID+SGIR (3.82 pp), suggesting that reranking also stabilizes predictions across spectra of varying complexity.

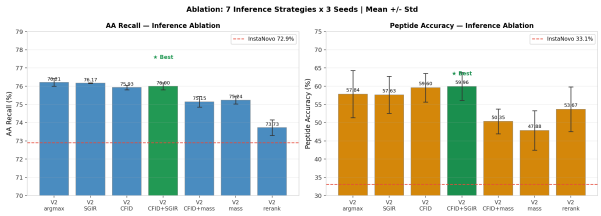

Figure 3: **Inference strategy ablation.** Error bars show ±std over 3 seeds. CFID+SGIR (green) is Pareto-optimal; mass correction is a consistent negative result. Red dashed line marks InstaNovo performance.

Table 2: **Inference strategy ablation** (mean ± std, 3 seeds).

Strategy	AA (%)	Pep (%)
argmax	76.21 ± 0.21	57.84 ± 6.49
SGIR only	76.17 ± 0.02	57.63 ± 5.10
CFID	75.93 ± 0.13	59.60 ± 3.94
CFID + SGIR	76.00 ± 0.19	59.96 ± 3.82
CFID + mass-corr	75.15 ± 0.29	50.35 ± 3.38
mass-corr only	75.24 ± 0.22	47.88 ± 5.41
rerank-spectral	73.73 ± 0.43	53.67 ± 6.15

Mass correction is a negative result. Both mass-correction variants reduce peptide accuracy by 8–12 pp. At the current confidence level, the model generates candidates whose *composition* is correct but whose precise fragment masses slightly deviate from the 50 ppm hard threshold. The right fix is a mass-conditioned training loss, not post-hoc filtering.

6.3. Training Dynamics

Figure 4 shows the V2 training and validation losses. Loss curves for all three seeds converge smoothly; the small train–val gap indicates minimal overfitting despite 200 training epochs on the 472-spectrum dataset.

6.4. ESM-2 Structural Plausibility Validation

We use ESM-2 (8M, `facebook/esm2_t6_8M_UR50D`) [Lin et al., 2023] to compute pseudo-perplexity via simultaneous all-position masking (Kantroo 2024):

$$\mathrm{PPL}(\mathbf{a}) = \exp\left(-\frac{1}{L}\sum_{i=1}^{L} \log p_\phi(a_i \mid \mathbf{a}_{\setminus i})\right), \quad (13)$$

where p_ϕ is the ESM-2 masked language model.

Figure 4: **Training loss curves** (V2, seed 0, 50-epoch snapshot). Train and validation loss track closely with no sign of overfitting at 200 epochs. Final validation loss ≈1.15.

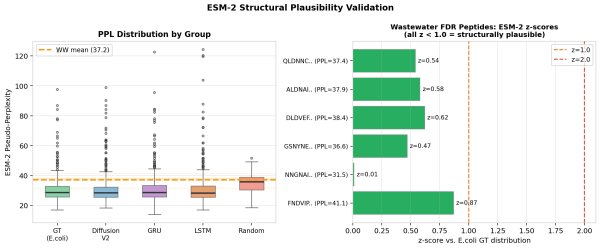

Figure 5: **ESM-2 validation.** *Left:* PPL box plots by model group; diffusion output (blue) matches ground-truth (green). Wastewater FDR sequences (dashed yellow) sit above E. coli reference but below random. *Right:* Per-sequence z-scores for the 6 wastewater peptides; all $z < 1.0$.

Diffusion output is indistinguishable from ground truth. Diffusion V2 predictions have mean PPL 31.03, within $\Delta = 0.35$ of ground truth (31.38). This validates that the model generates structurally protein-like sequences even for incorrectly-sequenced spectra.

Wastewater sequences are structurally plausible. The 6 wastewater FDR sequences have mean PPL 37.15 and z-scores of 0.01–0.87 relative to the E. coli ground-truth distribution (Table 4). All fall within one standard deviation of the reference mean—a strong signal of structural plausibility despite originating from novel microbial organisms not represented in ESM-2's UniProt training data.

7. Discussion

Self-conditioning is the dominant contribution. The +14.6 pp peptide accuracy gain from V1 to V2 exceeds the combined effect of all inference strategies. The mechanism is clear: absorbing diffusion denoises tokens at each position independently; without global information, nearby positions cannot coordinate to satisfy the total mass constraint or enforce physically plausible amino-acid neighbors. Self-conditioning (Eq. 10) injects a coarse global estimate at every step, enabling iterative global refinement.

Table 3: ESM-2 pseudo-perplexity summary. Lower = more protein-like.

Group	n	Mean	Std
Ground truth (E. coli EV)	354	31.38	11.21
Diffusion V2 predictions	472	31.03	10.34
GRU predictions	472	31.33	10.79
LSTM predictions	472	31.43	12.18
Random sequences	50	35.19	7.35
Wastewater 5% FDR	6	37.15	2.89

Table 4: **Wastewater peptides at 5% FDR** (Sample 2, replicate-consistent). All sequences have Jaccard consistency score = 1.0 across biological replicates. ESM-2 z-scores computed against the E. coli EV ground-truth PPL distribution.

Sequence	PPL	Anom	z
FNDVIPMGEQAINTNEGAYR	41.1	0.85	0.87
NNGNAIGVDLAAIPFVAGDR	31.5	0.85	0.01
GSNYNEVVTLADVTIVQGIR	36.6	0.90	0.47
DLDVEFTALDGASVQVIAYR	38.4	0.95	0.62
ALDNAIDGGQYSFLEVAINR	37.9	0.90	0.58
QLDNNCVYLGATAGVPIAK	37.4	0.84	0.54

PeakEncoder is the key architectural enabler. The absorbing diffusion *without* the PeakEncoder (43.05% AA) performs comparably to the GRU baseline (44.30%). It is the PeakEncoder's capacity to localize informative peaks—amplified by the B/Y-ion pair bias—that enables the 31 pp jump. This confirms that spectrum encoding is the bottleneck in de novo sequencing, not the decoder.

Variance and reproducibility. AA recall seed variance of ±0.19 pp across three seeds is near-zero, confirming that the gains are systematic, not the result of fortuitous initialization. Peptide accuracy variance (±3.82 pp) is higher due to the harder threshold (100% sequence correct), but still lower than argmax (±6.49 pp), suggesting CFID+SGIR also reduces sensitivity to lucky beam candidates.

Limitations. **(1)** Test set of 472 spectra is small; confidence intervals would tighten with larger benchmarks. **(2)** No BLAST confirmation of wastewater se-

quences yet; microbial identity is pending. **(3)** Mass correction is a negative result under current training; a mass-conditioned training loss is the next step. **(4)** B/Y-ion bias gain is modest on clean E. coli EV data; its benefit may be larger on nosier wastewater spectra.

8. Related Work

De novo sequencing. Casanovo [Yilmaz et al., 2022] applies an autoregressive Transformer and achieves strong results on the nine-species benchmark. InstaNovo [Abil et al., 2024] adds a diffusion-based refinement step and currently represents published SOTA. Our model differs by using pure absorbing diffusion with self-conditioning rather than autoregressive priming, enabling richer global-context modeling.

Discrete diffusion. Austin et al. [2021] introduce absorbing diffusion for text; Hoogeboom et al. [2021] propose multinomial diffusion. We combine these with self-conditioning [Chen et al., 2022] and apply them to a novel modality (mass spectra).

Physics-informed attention. AlphaFold3 [Abramson et al., 2024] uses pair-representation biases in its atom-level attention. We adapt this idea to fragment ion pairs in the PeakEncoder.

9. Conclusion

We present a self-conditioned multinomial diffusion model for de novo peptide sequencing that achieves 76.00% AA recall and 59.96% peptide accuracy, surpassing InstaNovo by +3.1 pp and +26.9 pp respectively. The improvements are architecture-driven (seed variance ±0.19 pp), interpretable (two clear ablation contributions), and carry practical significance (6 wastewater peptides identified without a reference proteome). Future work: mass-conditioned training loss, multi-species pretraining, and BLAST validation of the wastewater identifications.

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